

Skills Summary

Estimating Differences and Checking Reasonableness

Use this reference while completing your worksheet or when studying.

1. Estimate a difference by rounding

The rule: round *both* numbers to the same place, then subtract the rounded numbers. Write $\approx$ (“is about”), not $=$.

Nearest hundred: look at the *tens* digit — 5 or more rounds up, less than 5 rounds down.

Nearest ten: look at the *ones* digit, same rule.

Rounding only one of the two numbers is the most common mistake; the estimate then drifts far from the real difference.

Example: Estimate $583 - 216$ to the nearest hundred.

Step 1. The question asks about how many, not exactly how many, so round each number first and subtract the easy numbers instead of regrouping.

Step 2. $583 \rightarrow 600$ (tens digit 8); $216 \rightarrow 200$ (tens digit 1).

Step 3. $600 - 200 = 400$, so the exact difference should land near 400.

Try it: Estimate $742 - 385$ to the nearest hundred.

check: 300

2. Is my exact answer reasonable?

The rule: the estimate is a *safety net*. Find the exact difference, then hold it next to the estimate.

Close to the estimate $\Rightarrow$ the answer is reasonable.

Far from the estimate $\Rightarrow$ a trade was probably missed — go back and rework the subtraction before you write the answer down.

“Close” means a few tens away when you rounded to the nearest hundred.

Example: A jar holds 405 beads. 168 beads are taken out. How many beads are left, and is the answer reasonable?

Step 1. Estimate first so there is something to compare the exact answer with, then subtract exactly.

Step 2. Estimate: $400 - 200 = 200$ beads. Exact: trade across the zero — 405 becomes 3 hundreds, 9 tens, 15 ones; $15 - 8 = 7$, $9 - 6 = 3$, $3 - 1 = 2$.

Step 3. The exact answer **237** beads sits close to the estimate of 200 beads, so it is reasonable.

Try it: 612 marbles, 289 given away. Find the exact number left and check it against your estimate.

check: 323

3. Check a difference by adding

The rule: addition undoes subtraction.

If $\text{start} - \text{taken away} = \text{answer}$, then $\text{answer} + \text{taken away} = \text{start}$.

Add your answer to the number you subtracted. Getting the starting number back proves every trade was right; getting anything else proves it was not.

Example: Is $486 - 259 = 227$ correct?

Step 1. Checking a difference means running it backwards, so add the answer to the number that was taken away.

Step 2. $227 + 259$: $7 + 9 = 16$ (write 6, carry 1); $2 + 5 + 1 = 8$; $2 + 2 = 4$.

Step 3. The check gives **486**, the number we started from, so 227 is correct.

Try it: Check $350 - 176 = 174$ by adding $174 + 176$.

check:

(!) Watch out: When rounding to the nearest hundred, look at the *tens* digit, not the ones digit. 349 rounds *down* to 300 even though it ends in 9.